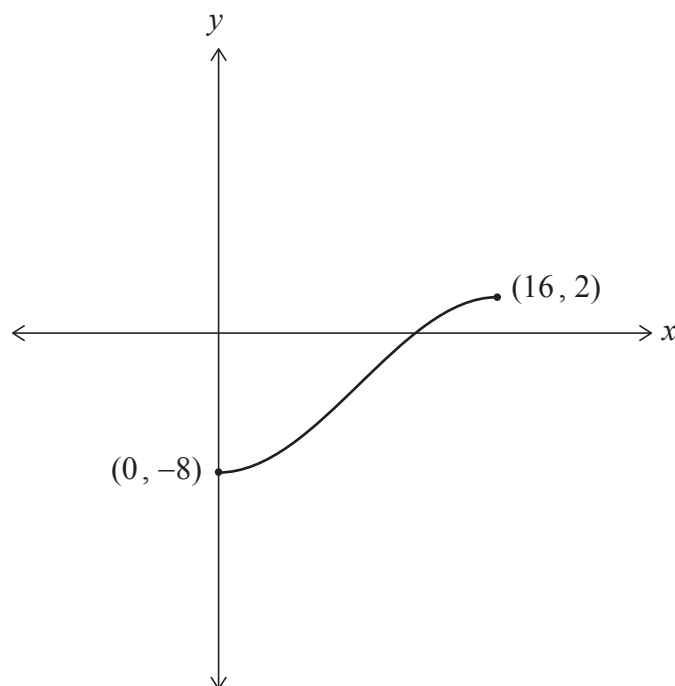


4. [Maximum mark: 7]

Consider the function $g(x) = a \cos(bx) - 3$, where $x \in \mathbb{R}$ and $a, b \in \mathbb{R}$.

The following diagram shows part of the graph of g .



The graph of g has a local minimum at $(0, -8)$ and a local maximum at $(16, 2)$.

(a) Find the value of

(i) a ;

(ii) b , where $b > 0$.

[3]

(b) Write down the smallest positive value of the constant k such that $g(x + k) = g(x)$ for all x .

[1]

(c) The function $g(x)$ can be written in the form $f(x) = p \sin b(x - q) - 3$ where $p, q \in \mathbb{Z}^+$.

(i) Find the smallest positive value of q .

(ii) For this value of q , write down the value of p .

[3]

(This question continues on the following page)



(Question 4 continued)

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